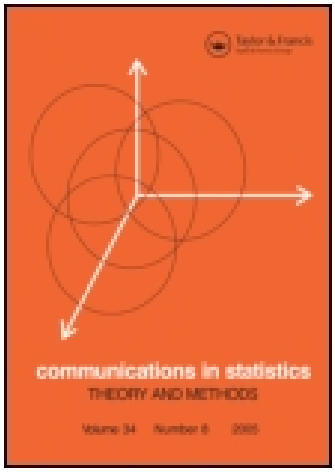
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Robustness of bayesian factor analysis estimates

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ROBUSTNESS OF BAYESIAN FACTOR ANALYSIS ESTIMATES

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Key Words: Bayes factor analysis, ϵ - contamination prior class, directional derivatives, robustness

ABSTRACT

This paper presents the result of a study of the robustness of posterior estimators of the factor loading matrix, the factor scores, and the disturbance covariance matrix(the main model parameters) in a Bayesian factor analysis with respect to variations in the values of the parameters of their prior distributions(the hyperparameter). We adopt the ϵ - contamination model of Berger and Berliner(1986) to generate prior distributions whose hyperparamters reflects small variations in the elements of the uncontaminated hyperparameters, and we use directional derivatives to examine the variation of the uncontaminated estimators with respect to changes in the values of the hyperparameters, in the directions of the main model parameters. Several matrix norms are used to measure the closeness of the resulting values. We illustrate the results with a numerical example.

1. INTRODUCTION

A Baysian approach to the classical factor analysis model was proposed in Press, 1972 based on the use of vague prior distributions. Kaufman and Press, 1976 examined a modified factor analysis model that contained more latent factors than observational dimensions(see also, Guttman, 1953).

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Heywood cases were examined by Martin and McDonald, 1975, who obtained implicit numerical solutions under a vague prior for the classical model. Wong, 1980 reexamined the problem from the empirical Bayesian point of view using normal priors for the factor loadings, and using the EM algorithm to obtain an implicit solution. Lee, 1981 did a hierarchical Bayesian analysis and obtained an implicit solution. Some other contributors to the Bayesian factor analysis problem include Euverman and Vermulst, 1983, Mayekawa, 1985, and Shigemasu, 1986. These analysis generally examined the classical model, and treated factor loadings and disturbance covariance as separate from the factor scores(except for the Mayekawa study in the latter case). Moreover, estimation solutions were iterative, but provided point estimators.

The Bayesian factor analysis problem was reconsidered more recently by Press and Shigemasu, 1989 and 1997 (hereafter referred to as PS). They proposed that we adopt a variant of the classical model, one which uses a disturbance covariance matrix that is general, instead of diagonal, but whose expected values is diagonal, a priori. They then estimated the factor loading matrix, the factor scores, and the disturbance covariance matrix of this model jointly from the posterior distribution. Their solution is non-iterative, and provides point as well as interval estimators; in fact, a complete posterior distribution results, making it a straightforward matter to compute credibility intervals, test hypotheses, or make decisions.

It is the PS model whose robustness to hyperparameter changes is examined in the present paper. Gustafson, 1996, adopted an approach

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to Bayesian robustness in which a posterior expectation was treated as a function of the prior marginal, and the derivative of the functional was evaluated at the nominal prior marginal. This approach automatically provides a convenient measure of sensitivity. By contrast, in the Berger/Berliner, 1986, approach adopted here, it is necessary to devise robustness measures of sensitivity to fit the problem (as will be done later), but the approach has the advantage that it is not necessary to view the problem in terms of marginals.

The plan of this paper is as follows. Section 2 sets out the basic sampling model under consideration. Section 3 explicates the contamination robustness model we've adopted. Section 4 develops the formulas for the joint and marginal posterior densities of the unknown quantities in the contamination robustness model, while in Section 5, we give a point estimators of the factor loading matrix, the factor scores, and the disturbance covariance matrix in terms of the contamination parameters. In section 6 we check the estimators for the robustness by evaluating the directional derivatives of the matrix estimators with respect to the hyperparameters. We conclude in Sect. 7 with a numerical example.

2 THE PS MODEL

Let :

x_i denote a (px1) vectors of observations for subject i centered at their mean vector, so that $E(x_i)$=0; X=$(x_1, ..., x_N)'$;

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Λ denote the (pxm) factor loading matrix, m denotes the number of factors; m <= p;

f_i denotes the (mx1) factor score vector for subject i; the factor score matrix is
F=($f_1, ..., f_N$);

ϵ_i denotes the (px1) vector of disturbances, i=1,...,N.

Adopt the model :

$$(x_i|\Lambda, f_i) = \Lambda f_i + \epsilon_i \qquad i = 1, ..N$$

with

$$\epsilon_i \sim N(0, \Psi).$$

Note that $\Psi > 0$ implies that the error covariance matrix may be a full, non-diagonal positive definite matrix. The likelihood function is then given by

$$P(X|\Lambda, F, \Psi) \propto |\Psi|^{-\frac{N}{2}} exp\frac{-1}{2} tr\Psi^{-1}(X - F\Lambda')(X - F\Lambda')' \quad (2.1)$$

Adopt the prior density structure for the unknown model parameters,

$$P(\Lambda, F, \Psi) = P(\Lambda|\Psi)P(\Psi)P(f) \quad (2.2)$$

We take the prior for $(\Lambda|\Psi)$ to be matrix normal, and that for Ψ to be inverted Wishart.

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Thus , take:

$$P(\Lambda|\Psi) \propto |\Psi|^{-\frac{m}{2}} exp\frac{-1}{2} tr(\Lambda - \Lambda_0)H(\Lambda - \Lambda_0)'\Psi^{-1}, \quad (2.3)$$

$$P(\Psi) \propto |\Psi|^{-\frac{\nu}{2}} exp\frac{-1}{2} tr\Psi^{-1}B \quad (2.4)$$

Now assume, in keeping with traditional views of the factor model containing "common" and "specific" factors, that B is a diagonal matrix, so that E($\Psi|B$) is diagonal.

3 THE ϵ-CONTAMINATION MODEL FOR ROBUSTNESS

The ϵ-contamination model for evaluating model robustness was introduced by Berger and Berliner(1986). The context is that we are assuming a "base" prior distribution, say $p(\theta)$. We are concerned about how our inferences may be affected by priors that are somewhat different from $p(\theta)$, by small amounts. They suggested that if we wish to evaluate the effect on posterior inference of a small ϵ-variation of the prior distribution away from what we have assumed we should study the behavior of a convex mixture of the base prior and contamination densities. We will adopt this approach to evaluating the robustness of the PS Bayesian factor analysis model by considering contamination models for the prior distribution that have the same form as that of the original prior, but with different hyperparameters. Thus, we'll be addressing the issue of how posterior inferences are affected by small variations in the assessed values of the parameters of the prior distribution.

For example, suppose that a priori, for the some unknown quantity θ,

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$\theta \sim N(\psi, \sigma^2)$. To study robustness, we might consider the ϵ-contamination model prior :

$$f(\theta) = (1-\epsilon)p(\theta|\psi,\sigma^2) + (\epsilon)p(\theta|\psi_0,\sigma_0), \tag{3.5}$$

where p(.) denotes the normal density function.

4 POSTERIOR DISTRIBUTION FOR THE PARAMETERS OF THE ROBUSTNESS CONTAMINATION MODEL

By Bayes' theorem,

$$P(\Lambda, F, \Psi|X) \propto P(X|\Lambda, F, \Psi)P(\Lambda|\Psi)P(\Psi)P(F) \tag{4.6}$$

Substituting (2.1)-(2.5) into (4.7), but replacing the priors by convex mixtures of base and contamination priors, as in (3.6), gives

$$P(\Lambda, F, \Psi|X) \propto P(X|\Lambda, F, \Psi)P(\Lambda, F, \Psi) \tag{4.7}$$

$$\propto P(X|\Lambda, F, \Psi)P(\Lambda|\Psi)P(\Psi)P(F) \tag{4.8}$$

$$\propto [(1-\epsilon)^2|\Psi|^{-(N+m+\upsilon)/2}exp(-1/2)tr\Psi^{-1}G] + \tag{4.9}$$
$$[(1-\epsilon)(\epsilon)|\Psi|^{-(N+m_1+\upsilon)/2}exp(-1/2)tr\Psi^{-1}G_1] +$$
$$[(1-\epsilon)(\epsilon)|\Psi|^{-(N+m_1+\upsilon_1)/2}exp(-1/2)tr\Psi^{-1}G_2] +$$
$$(\epsilon)^2|\Psi|^{-(N+m+\upsilon_1)/2}exp(-1/2)tr\Psi^{-1}G_3]P(F)$$

where

$$G = (X - F\Lambda')'(X - F\Lambda') + (\Lambda - \Lambda_0)H(\Lambda - \Lambda_0)' + B$$
$$G_1 = (X - F\Lambda')'(X - F\Lambda') + (\Lambda - \Lambda_0)H(\Lambda - \Lambda_0)' + B_1$$
$$G_2 = (X - F\Lambda')'(X - F\Lambda') + (\Lambda - \Lambda_1)H_1(\Lambda - \Lambda_1)' + B$$
$$G_3 = (X - F\Lambda')'(X - F\Lambda') + (\Lambda - \Lambda_1)H_1(\Lambda - \Lambda_1)' + B_1$$

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MARGINAL POSTERIOR

$$P(\Lambda, F|X) \propto \int P(\Lambda, F, \Psi|X)\partial\Psi$$

Using the properties of the inverted Wishart density

$$\begin{aligned} P(\Lambda, F|X) \propto\ & (1-\epsilon)^2|G|^{-(N+m+v)/2} + \\ & (1-\epsilon)(\epsilon)|G_1|^{-(N+m+v_1)/2} + \\ & (\epsilon)(1-\epsilon)|G_2|^{-(N+m_1+v)/2} + \\ & (\epsilon)^2|G_3|^{-(N+m_1+v_1)/2} \end{aligned} \tag{4.10}$$

Before we try to determine P(F|X) which is found by integrating(4.11) with respect Λ, a form which is proportional to a matrix T density, we expand the G's and complete the square in Λ.

Thus

$$\begin{aligned} P(\Lambda, F|X) \propto\ & [(1-\epsilon)^2|R_F + (\Lambda-\Lambda_F)Q_F(\Lambda-\Lambda_F)|^{-r/2} + \\ & (1-\epsilon)(\epsilon)|R_{F_1} + (\Lambda-\Lambda_F)Q_F(\Lambda-\Lambda_F)|^{-r_1/2} + \\ & (1-\epsilon)(\epsilon)|R_{F_2} + (\Lambda-\Lambda_{F_1})Q_{F_1}(\Lambda-\Lambda_{F_1}|^{-r_2/2} + \\ & (\epsilon)^2|R_{F_3} + (\Lambda-\Lambda_{F_1})Q_{F_1}(\Lambda-\Lambda_{F_1})|^{-r_3/2}]P(F) \end{aligned} \tag{4.11}$$

where

$$Q_F = H + F'F$$

$$Q_{F_1} = H_1 + F'F$$

$$R_F = X'X + \Lambda_0 H \Lambda_0' + B - (X'F + \Lambda_0 H)Q_F^{-1}(X'F + \Lambda_0 H)'$$

$$R_{F_1} = X'X + \Lambda_0 H \Lambda_0' + B_1 - (X'F + \Lambda_0 H)Q_F^{-1}(X'F + \Lambda_0 H)'$$

$$R_{F_2} = X'X + \Lambda_1 H \Lambda_1' + B - (X'F + \Lambda_1 H_1)Q_{F_1}^{-1}(X'F + \Lambda_1 H_1)'$$

$$R_{F_3} = X'X + \Lambda_1 H \Lambda_1' + B_1 - (X'F + \Lambda_1 H_1)Q_{F_1}^{-1}(X'F + \Lambda_1 H_1)'$$

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$$\Lambda_F = (X'F + \Lambda_0 H)Q_F^{-1}$$

$$\Lambda_{F_1} = (X'F + \Lambda_1 H_1)Q_{F_1}^{-1}$$

$$r = N + m + v - p - 1$$

$$r_1 = N + m + v_1 - p - 1$$

$$r_2 = N + m_1 + v - p - 1$$

$$r_3 = N + m_1 + v_1 - p - 1$$

Using the normalizing constant of a matrix T-distribution, we can integrate $P(\Lambda, F|X)$ with respect to Λ

This gives for the marginal posterior density of F given X,

$$\begin{aligned} P(F|X) \propto\ & (1-\epsilon)^2 \frac{P(F)}{|R_F|^{(r-m)/2}|Q_F|^{p/2}} + \\ & (1-\epsilon)(\epsilon) \frac{P(F)}{|R_{F_1}|^{(r_1-m)/2}|Q_F|^{p/2}} + \\ & (1-\epsilon)(\epsilon) \frac{P(F)}{|R_{F_2}|^{(r_2-m_1)/2}|Q_{F_1}|^{p/2}} + \\ & (\epsilon)^2 \frac{P(F)}{|R_{F_3}|^{(r_3-m_1)/2}|Q_{F_1}|^{p/2}} \end{aligned} \tag{4.12}$$

$$\propto\ P_1(F) + \epsilon P_2(F) + \epsilon^2 P_3(F) \tag{4.13}$$

where

$$P_1(F) = \frac{P(F)}{|R_F|^{(r-m)/2}|Q_F|^{p/2}} \tag{4.14}$$

$$P_2(F) = \frac{P(F)}{|R_{F_1}|^{(r_1-m)/2}|Q_F|^{p/2}} + \frac{P(F)}{|R_{F_2}|^{(r_2-m_1)/2}|Q_{F_1}|^{p/2}} - \frac{2P(F)}{|R_F|^{(r-m)/2}|Q_F|^{p/2}} \tag{4.15}$$

$$P_3(F) = \frac{P(F)}{|R_F|^{(r-m)/2}|Q_F|^{p/2}} - \frac{P(F)}{|R_{F_1}|^{(r_1-m)/2}|Q_F|^{p/2}} - \tag{4.16}$$

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$$\frac{P(F)}{|R_{F_2}|^{(r_2-m_1)/2}|Q_{F_1}|^{p/2}} + \frac{P(F)}{|R_{F_3}|^{(r_3-m_1)/2}|Q_{F_1}|^{p/2}}$$

Futhermore, the marginal posterior density of F can be rewritten in the form

$$P(F|X) \propto P_1(F) + \epsilon P_2(F) + \epsilon^2 P_3(F)$$

with

$$P_1(F) = \frac{P(F)|H + F'F|^{(r-m-p)/2}}{|A - (F - \hat{F}_0)'(I_N - XW^{-1}X')(F - \hat{F}_0)'|^{(r-m)/2}} \quad (4.17)$$

$$P_2(F) = \frac{P(F)|H + F'F|^{(r_1 m-p)/2}}{|A_1 - (F - \hat{F}_1)'(I_N - XW_1^{-1}X')(F - \hat{F}_1)'|^{(r_1-m)/2}} + \frac{P(F)|H_1 + F'F|^{(r_2-m_1-p)/2}}{|A_2 - (F - \hat{F}_2)'(I_N - XW_2^{-1}X')(F - \hat{F}_2)'|^{(r_2-m_1)/2}} - \frac{2P(F)|H + F'F|^{(r-m-p)/2}}{|A - (F - \hat{F}_0)'(I_N - XW^{-1}X')(F - \hat{F}_0)'|^{(r-m)/2}} \quad (4.18)$$

$$P_3(F) = \frac{P(F)|H + F'F|^{(r-m-p)/2}}{|A - (F - \hat{F}_0)'(I_N - XW^{-1}X')(F - \hat{F}_0)'|^{(r-m)/2}} - \frac{P(F)|H + F'F|^{(r_1 m-p)/2}}{|A_1 - (F - \hat{F}_1)'(I_N - XW_1^{-1}X')(F - \hat{F}_1)'|^{(r_1-m)/2}} - \frac{P(F)|H_1 + F'F|^{(r_2-m_1-p)/2}}{|A_2 - (F - \hat{F}_2)'(I_N - XW_2^{-1}X')(F - \hat{F}_2)'|^{(r_2-m_1)/2}} + \frac{P(F)|H_1 + F'F|^{(r_3-m_1-p)/2}}{|A_3 - (F - \hat{F}_3)'(I_N - XW_3^{-1}X')(F - \hat{F}_3)'|^{(r_3-m_1)/2}} \quad (4.19)$$

where

$$\hat{F}_0 = (I_N - XW^{-1}X')^{-1}XW^{-1}\Lambda_0 H$$

$$\hat{F}_1 = (I_N - XW_1^{-1}X')^{-1}XW_1^{-1}\Lambda_0 H$$

$$\hat{F}_2 = (I_N - XW_2^{-1}X')^{-1}XW_2^{-1}\Lambda_1 H_1$$

$$\hat{F}_3 = (I_N - XW_3^{-1}X')^{-1}XW_3^{-1}\Lambda_1 H_1$$

$$W = X'X + B + \Lambda_0 H \Lambda_0'$$

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$$W_1 = X'X + B_1 + \Lambda_0 H \Lambda_0'$$

$$W_2 = X'X + B + \Lambda_1 H_1 \Lambda_1'$$

$$W_3 = X'X + B_1 + \Lambda_1 H_1 \Lambda_1'$$

$$A = H - H'\Lambda_0' W^{-1} \Lambda_0 H - (H'\Lambda_0' W^{-1} X')(I_N - XW^{-1}X')(XW^{-1}\Lambda_0 H)$$

$$A_1 = H - H'\Lambda_0' W_1^{-1} \Lambda_0 H - (H'\Lambda_0 W_1^{-1} X')(I_N - XW_1^{-1}X')(XW_1^{-1}\Lambda_0 H)$$

$$A_2 = H_1 - H_1'\Lambda_1' W_2^{-1} \Lambda_1 H_1 - (H_1'\Lambda_1 W_2^{-1} X')(I_N - XW_2^{-1}X')(XW_2^{-1}\Lambda_1 H_1)$$

$$A_3 = H_1 - H_1'\Lambda_1' W_3^{-1} \Lambda_1 H_1 - (H_1'\Lambda_1 W_3^{-1} X')(I_N - XW_3^{-1}X')(XW_3^{-1}\Lambda_1 H_1)$$

5 ESTIMATES OF F, Λ, Ψ

In this section we develop Bayesian estimates(F, Λ, Ψ) using the ϵ-contamination class defined above.

Estimation of the Factor Score: $\hat{F}$

We adopt the vague prior for F,

$$P(F) \propto constant$$

Equation (17)-(19) become

$$P_1(F) \propto \frac{|H + F'F|^{(r-m-p)/2}}{|A - (F - \hat{F}_0)'(I_N - XW^{-1}X')(F - \hat{F}_0)'|^{(r-m)/2}} \tag{5.20}$$

$$P_2(F) \propto \frac{|H + F'F|^{(r_1 m - p)/2}}{|A_1 - (F - \hat{F}_1)'(I_N - XW_1^{-1}X')(F - \hat{F}_1)'|^{(r_1 - m)/2}} + \frac{|H_1 + F'F|^{(r_2 - m_1 - p)/2}}{|A_2 - (F - \hat{F}_2)'(I_N - XW_2^{-1}X')(F - \hat{F}_2)'|^{(r_2 - m_1)/2}} - \frac{2|H + F'F|^{(r-m-p)/2}}{|A - (F - \hat{F}_0)'(I_N - XW^{-1}X')(F - \hat{F}_0)'|^{(r-m)/2}} \tag{5.21}$$

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$$P_3(F) \propto \frac{|H+F'F|^{(r-m-p)/2}}{|A-(F-\hat{F}_0)'(I_N-XW^{-1}X')(F-\hat{F}_0)'|^{(r-m)/2}} - \frac{|H+F'F|^{(r_1m-p)/2}}{|A_1-(F-\hat{F}_1)'(I_N-XW_1^{-1}X')(F-\hat{F}_1)'|^{(r_1-m)/2}} - \frac{|H_1+F'F|^{(r_2-m_1-p)/2}}{|A_2-(F-\hat{F}_2)'(I_N-XW_2^{-1}X')(F-\hat{F}_2)'|^{(r_2-m_1)/2}} + \frac{|H_1+F'F|^{(r_3-m_1-p)/2}}{|A_3-(F-\hat{F}_3)'(I_N-XW_3^{-1}X')(F-\hat{F}_3)'|^{(r_3-m_1)/2}} \quad (5.22)$$

Recall that $\frac{1}{N}F'F = \frac{1}{N}\sum_i^N f_j f_j'$, so that by the law of large of numbers, for large N, we have approximately that $\frac{1}{N}F'F \sim I$. So that |H+F'F| in (5.21)-(5.23) becomes approximately constant(not depedent upon F). Then for large N, P(F|X) becomes a matrix T-distribution with density,

$P(F|X) \propto P_1(F) + \epsilon P_2(F) + \epsilon^2 P_3(F)$

with

$$P_1(F) \propto |A-(F-\hat{F}_0)'(I_N-XW^{-1}X')(F-\hat{F}_0)'|^{-(r-m)/2} \quad (5.23)$$

$$P_2(F) \propto |A_1-(F-\hat{F}_1)'(I_N-XW_1^{-1}X')(F-\hat{F}_1)'|^{-(r_1-m)/2} + |A_2-(F-\hat{F}_2)'(I_N-XW_2^{-1}X')(F-\hat{F}_2)'|^{-(r_2-m_1)/2} - 2|A-(F-\hat{F}_0)'(I_N-XW^{-1}X')(F-\hat{F}_0)'|^{-(r-m)/2} \quad (5.24)$$

$$P_3(F) \propto |A-(F-\hat{F}_0)'(I_N-XW^{-1}X')(F-\hat{F}_0)'|^{-(r-m)/2} - |A_1-(F-\hat{F}_1)'(I_N-XW_1^{-1}X')(F-\hat{F}_1)'|^{-(r_1-m)/2} - |A_2-(F-\hat{F}_2)'(I_N-XW_2^{-1}X')(F-\hat{F}_2)'|^{-(r_2-m_1)/2} + |A_3-(F-\hat{F}_3)'(I_N-XW_3^{-1}X')(F-\hat{F}_3)'|^{-(r_3-m_1)/2} \quad (5.25)$$

With squared error, $\hat{F}$ becomes the E(F|X).

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$$\hat{F} = \hat{F}_0 + \epsilon(\hat{F}_1 + \hat{F}_2 - 2\hat{F}_0) + \epsilon^2(\hat{F} - \hat{F}_1 - \hat{F}_2 + \hat{F}_3) \quad (5.26)$$

Estimation of the Factor Loading Matrix : $\hat{\Lambda}$

From the marginal posterior density of (Λ,F|X), we can estimate Λ for given F=$\hat{F}$,X.

The Bayesian point estimator of Λ is E($\Lambda|\hat{F}$,X) :

$$\hat{\Lambda} = \Lambda_{\hat{F}} + \epsilon(\Lambda_{\hat{F}_1} - \hat{\Lambda}_{\hat{F}}) \quad (5.27)$$

Estimation of Disturbance Covariance Matrix : $\hat{\Psi}$

The distribution of covariance matrix ,Ψ, is estimated, conditional upon ($\hat{\Lambda}, \hat{F}$) with the joint posterior density of (Λ, F,Ψ|X) given and

$$\begin{aligned}\hat{\Psi} = {} & \frac{\hat{G}}{N+m+\upsilon-2p-2} + \epsilon[\frac{\hat{G}_1}{N+m+\upsilon_1-2p-2} + \\ & \frac{\hat{G}_2}{N+m_1-\upsilon-2p-2} - \frac{2\hat{G}}{N+m+\upsilon-2p-2}] + \\ & \epsilon^2[\frac{\hat{G}}{N+m+\upsilon-2p-2} - \frac{\hat{G}_1}{N+m+\upsilon_1-2p-2} - \\ & \frac{\hat{G}_2}{N+m+\upsilon-2p-2} + \frac{\hat{G}_3}{N+m_1+\upsilon_1-2p-2}]\end{aligned} \quad (5.28)$$

6 CHECKING FOR THE ROBUSTNESS

We have the point estimators, $\hat{F}, \hat{\Lambda}, \hat{\Psi}$. Checking for robustness, we would like to compare each estimator with one that results after changing the hyperparameter. In other words, how would each estimator, $\hat{F}$, $\hat{\Lambda}$, $\hat{\Psi}$ vary after changing the location parameter, Λ_1 and scale parameters B_1, H_1 and ϵ in the ϵ- contamination prior class.

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We examine this question by evaluating differences found in directional derivatives, using two distinct measures of distance.

We first define the averages of the absolute values in the column of $\hat{F}, \hat{\Lambda}$.

Define

$\bar{\hat{F}}_{.j} = \frac{1}{N} \sum_{i=1}^{N} |\hat{F}_{ij}|$ j=1,..,m.

$\bar{\hat{\Lambda}}_{.j} = \frac{1}{N} \sum_{i=1}^{N} |\hat{\Lambda}_{ij}|$, j=1,..,m.

where $|*|$ is the absolute value of the * . Examining the differences in such averages of column vectors between the estimated values for the baseline prior and that for the modified, or contaminated prior yields a measure of robustness. For the multivariate model we are considering here, it is useful to compare several distinct distance measures for describing quantitatively how much a small changes in the underlying prior distribution changes the estimates. We consider two different distance measures: Norm Distance, and Mahalanobis Distance.

Define

$$Norm(\Delta\bar{\hat{F}}) = \sqrt{\Delta(\bar{\hat{F}})'\Delta(\bar{\hat{F}})} = ND,$$
$$Norm(\Delta\hat{F}) = \sqrt{|\Delta\hat{F}'\Delta\hat{F}|} = ND_1$$

where $|*|$ is the determinant of * . ND is the simple Euclidean distance between $\bar{\hat{F}}_{.j}$ and $\bar{\hat{F}}_{.j'}$ and ND_1 is also the simple Euclidean distance using the whole matrix($\hat{F}$) instead of only the vector($\bar{\hat{F}}$).

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Another way to measure the distance between the estimated F and estimated contaminated F is the Mahalanobis distance, D^2, which is given by

$$D_{ij}^2 = (\bar{\hat{F}}_i - \bar{\hat{F}}_j)' S_F^{-1} (\bar{\hat{F}}_i - \bar{\hat{F}}_j)$$

where S_F denotes the pooled covariance matrix. Comparing D_{ij}^2 with ND and ND_1, D_{ij}^2 has the advantage of explicitly accounting for any correlations that might exist among the variables.

Directional Derivatives

Directional derivatives which measures the rate of changing of the function f in any arbitrary direction, u. Given a function f:D$\in R^m \rightarrow$ R, a point p$\in$D, and a vector u$\in R^m$, the directional derivatives of f at point p in the direction of the nonzero vector u is defined by

$$\begin{aligned} \frac{\partial f}{\partial u}(p) &= \lim_{t \to 0} \frac{f(p+tu) - (p)}{t} \\ &= \frac{d}{dt} f(p+tu)|_{t=0} \end{aligned}$$

Derivatives $\frac{\partial f}{\partial u}(p)$ are called directional derivatives, and they measure the rate of change of f in the direction of the vector u.

Now we apply directional derivatives to generalize the rate of change of each of the estimates $\hat{F}, \hat{\Lambda}$ and $\hat{\Psi}$, with the direction of location, and with the hyperparameters, Λ_1, B_1 and H_1.

By definition of directional derivatives, we can express the rate of change in $\hat{F}$ with direction λ_u on as

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$$\frac{d}{dt}\hat{F}_{\Lambda_1} = \frac{d}{dt}\hat{F}(\Lambda_1 + t\lambda_u)|_{t=0} \tag{6.29}$$

Therefore we can interpret that $\frac{d}{dt}\hat{F}_{\Lambda_1}$ is total derivatives of f at Λ_1 in the direction of λ_u.

For example, rate of change in $\hat{F}$ with direction λ_u, $\frac{d}{dt}\hat{F}_{\Lambda_1}$ is given by

$$\frac{d}{dt}\hat{F}_{\Lambda_1} = \frac{d}{dt}\hat{F}(\Lambda_1 + t\lambda_u)|_{t=0} \tag{6.30}$$

Similar to the direction b_u, h_u, the rates of changes in $\hat{F}$ with direction b_u, h_u are $\frac{d}{dt}\hat{F}_{B_1}, \frac{d}{dt}\hat{F}_{H_1}$ respectively.

For the rate of changes in $\hat{\Lambda}$ and $\hat{\Psi}$, we apply the same idea as for $\hat{F}$. However, since $\hat{\Lambda}$ is a functions of $\hat{F}$ and $\hat{\Psi}$ is the function of $\hat{F}$ and $\hat{\Lambda}$, we need to apply the chain rule in for directional derivatives.

From the results for the directional derivatives, see Lee, 1994, Appendices A1-A7, we conclude the following:

1) in the rate of changes in $\hat{F}$, with direction $\lambda_u(\frac{d}{dt}\hat{F}_{\Lambda_1})$, we see quadratic functions of λ_u, however with directions b_u and h_u, we only see linear function of b_u and h_u.

2) in rate of change in $\hat{\Lambda}$ and $\hat{\Psi}$, we can easily conclude that the rate of change $\hat{\Lambda}$ and $\hat{\Psi}$ are effected by the rate of change in $\hat{F}$ and $\hat{\Lambda}$.

Therefore at this point, we can conclude that the estimators($\hat{F}, \hat{\Lambda}$ and $\hat{\Psi}$) are more sensitive to variations in Λ_0 which is the location parameter in prior distribution, than to variations in B or H.

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7 NUMERICAL EXAMPLE

We adopt the example used by PS, 1989.

There are 48 applicants for a job and they have been scored on 15 variables regarding their acceptability. The variables are:

1) From of letter of application
2) Appearance
3) Academic ability
4) Likeability
5) Self-confidence
6) Lucidity
7) Honesty
8) Salesmanship
9) Experience
10) Drive
11) Ambition
12) Grasp
13) Potential
14) Keenness to join
15) Suitability

The raw scores of applicants on these variables, measured on the same scale are given by Table 1.

PS carried out a principal components analysis on the data and found four common factors. They used those results to structure their hyperparameters. We now examine the robustness of those results in the light of our analysis above.

The prior factor loading matrix, Λ_0, was taken to be:

$$\lambda_{0.7} = \Lambda_0 = \begin{bmatrix} 0 & 0 & 0.7 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0.7 & 0 & 0 \\ 0 & 0 & 0 & 0.7 \\ 0.7 & 0 & 0 & 0 \\ 0.7 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.7 \\ 0.7 & 0 & 0 & 0 \\ 0 & 0 & 0.7 & 0 \\ 0.7 & 0 & 0 & 0 \\ 0.7 & 0 & 0 & 0 \\ 0.7 & 0 & 0 & 0 \\ 0.7 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0.7 & 0 & 0 \end{bmatrix}.$$

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Table 1: Raw scores of 48 applcants scaled on 15 variables

Person	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	6	7	2	5	8	7	8	8	3	8	9	7	5	7	10
2	9	10	5	8	10	9	9	10	5	9	9	8	8	8	10
3	7	8	3	6	9	8	9	7	4	9	9	8	6	8	10
4	5	6	8	5	6	5	9	2	8	4	5	8	7	6	5
5	6	8	8	8	4	5	9	3	8	5	5	8	8	7	7
6	7	7	7	6	8	7	10	5	9	6	5	8	6	6	6
7	9	9	8	8	8	8	8	8	10	8	10	8	9	8	10
8	9	9	9	8	9	9	8	8	10	9	10	9	9	9	10
9	9	9	7	8	8	8	8	5	9	8	10	9	9	9	10
10	4	7	10	2	10	10	7	10	3	10	10	10	9	3	10
11	4	7	10	0	10	8	3	9	5	9	10	8	10	2	5
12	4	7	10	4	10	10	7	8	2	8	8	10	10	3	7
13	6	9	8	10	5	4	9	4	4	4	5	4	7	6	8
14	8	9	8	9	6	3	8	2	5	2	6	6	7	5	6
15	4	8	8	7	5	4	10	2	7	5	3	6	6	4	6
16	6	9	6	7	8	9	8	9	8	8	7	6	8	6	10
17	8	7	7	7	9	5	8	6	6	7	8	6	6	7	8
18	6	8	8	4	8	8	6	4	3	3	6	7	2	6	4
19	6	7	8	4	7	8	5	4	4	2	6	8	3	5	4
20	4	8	7	8	8	9	10	5	2	6	7	9	8	8	9
21	3	8	6	8	8	8	10	5	3	6	7	8	8	5	8
22	9	8	7	8	9	10	10	10	3	10	8	10	8	10	8
23	7	10	7	9	9	9	10	10	3	9	9	10	9	10	8
24	9	8	7	10	8	10	10	10	2	9	7	9	9	10	8
25	6	9	7	7	4	5	9	3	2	4	4	4	4	5	4
26	7	8	7	8	5	4	8	2	3	4	5	6	5	5	6
27	2	10	7	9	8	9	10	5	3	5	6	7	6	4	5
28	6	3	5	3	5	3	5	0	0	3	3	0	0	5	0
29	4	3	4	3	3	0	0	0	0	4	4	0	0	5	0
30	4	6	5	6	9	4	10	3	1	3	3	2	2	7	3
31	5	5	4	7	8	4	10	3	2	5	5	3	4	8	3
32	3	3	5	7	7	9	10	3	2	5	3	7	5	5	2
33	2	3	5	7	7	9	10	3	2	2	3	6	4	5	2
34	3	4	6	4	3	3	8	1	1	3	3	3	2	5	2
35	6	7	4	3	3	0	9	0	1	0	2	3	1	5	3
36	9	8	5	5	6	6	8	2	2	2	4	5	6	6	3
37	4	9	6	4	10	8	8	9	1	3	9	7	5	3	2
38	4	9	6	6	9	9	7	9	1	2	10	8	5	5	2
39	10	6	9	10	9	10	10	10	10	10	8	10	10	10	10
40	10	6	9	10	9	10	10	10	10	10	10	10	10	10	10
41	10	7	8	0	2	1	2	0	10	2	0	3	0	0	10
42	10	3	8	0	2	1	2	0	10	2	0	3	0	0	10
43	3	4	9	8	2	4	5	3	6	2	1	3	3	3	8
44	7	7	7	6	9	8	8	6	8	8	10	8	8	6	5
45	9	6	10	9	7	7	10	2	1	5	5	7	8	4	5
46	9	8	10	10	7	9	10	3	1	5	7	9	9	4	4
47	0	7	10	3	5	0	10	0	0	2	2	0	0	0	0
48	0	6	10	1	5	0	10	0	0	2	2	0	0	0	0

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The hyperparameter H was assessed as H=$10I_4$. The prior distribution for Ψ was assessed with B=$0.2I_{15}$ and υ=33.

Test for Robustness

We considered 6 cases :

Case 1

We fix B = $B_1 = 0.2I_{15}$

H = $H_1 = 10I_4$

$\epsilon = 0.1$ and $\Lambda_0 = \lambda_{0.7}$ and

let Λ_1 vary from $\lambda_{0.5}$ to $\lambda_{0.9}$

Case 2

We fix B = $B_1 = 0.2I_{15}$

H = $H_1 = 10I_4$

$\Lambda_0 = \lambda_{0.7}$ & $\Lambda_1 = \lambda_{0.8}$ and

let ϵ vary from 0.1 to 0.5

Case 3

We fix $\Lambda_0 = \Lambda_1 = \lambda_{0.7}$

H = $H_1 = 10\ I_4$

$\epsilon = 0.1$ & B = $0.2I_{15}$ and

let B_1 vary from $0.1I_{15}$ to $0.5I_{15}$

Case 4

We fix $\Lambda = \Lambda_0 = \lambda_{0.7}$

H = $H_1 = 10\ I_4$

B = $0.2I_{15}$&$B_1 = 0.1I_{15}$ and

let ϵ vary from 0.1 to 0.5

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Table 2: Robustness Summary of Cases

		Sensitive Cases		
Distance	Measure	F	Γ	Φ
Absolute	Value	case 1 case 2	case 1 case 2	
Mahalanobis	Distance		case 1 case 2	
Norm Distance	ND		case 1 case 2	
	ND1	case 1 case 2		
Directional Derivatives	LND		case 1	
	LND1	case 1		

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Case 5

We fix $\Lambda_0 = \Lambda_1 = \lambda_{0.7}$

$B = B_1 = 0.2I_{15}$

$\epsilon = 0.1$ and $H = 10I_4$ and

let H_1 vary from $8I_4$ to $12I_4$

Case 6

We fix $\Lambda_0 = \Lambda_1 = \lambda_{0.7}$

$B = B_1 = 0.2I_{15}$

$H_1 = 8I_4$ and $H = 10I_4$ and

let ϵ vary from 0.1 to 0.5

Robustness results are summarized in Table 2.

In cases 1, 3 and 5, we are testing the robustness in Λ_1, B_1 and H_1 respectively. In case 2, 4 and 6, we exame how the estimates are effected by contamination itself.

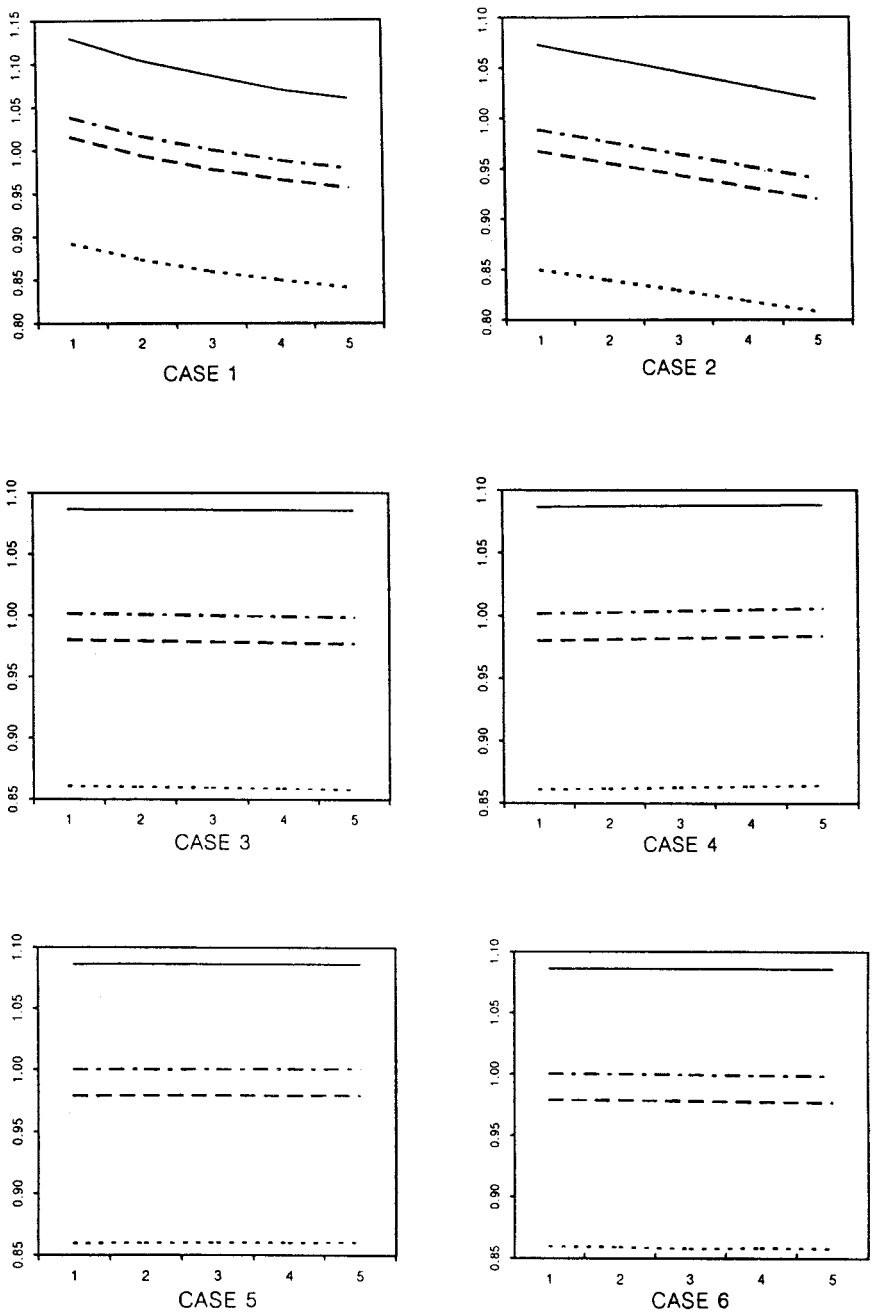


Figure 1: Absolute value; graphs of $\bar{\hat{F}}_{.j}$

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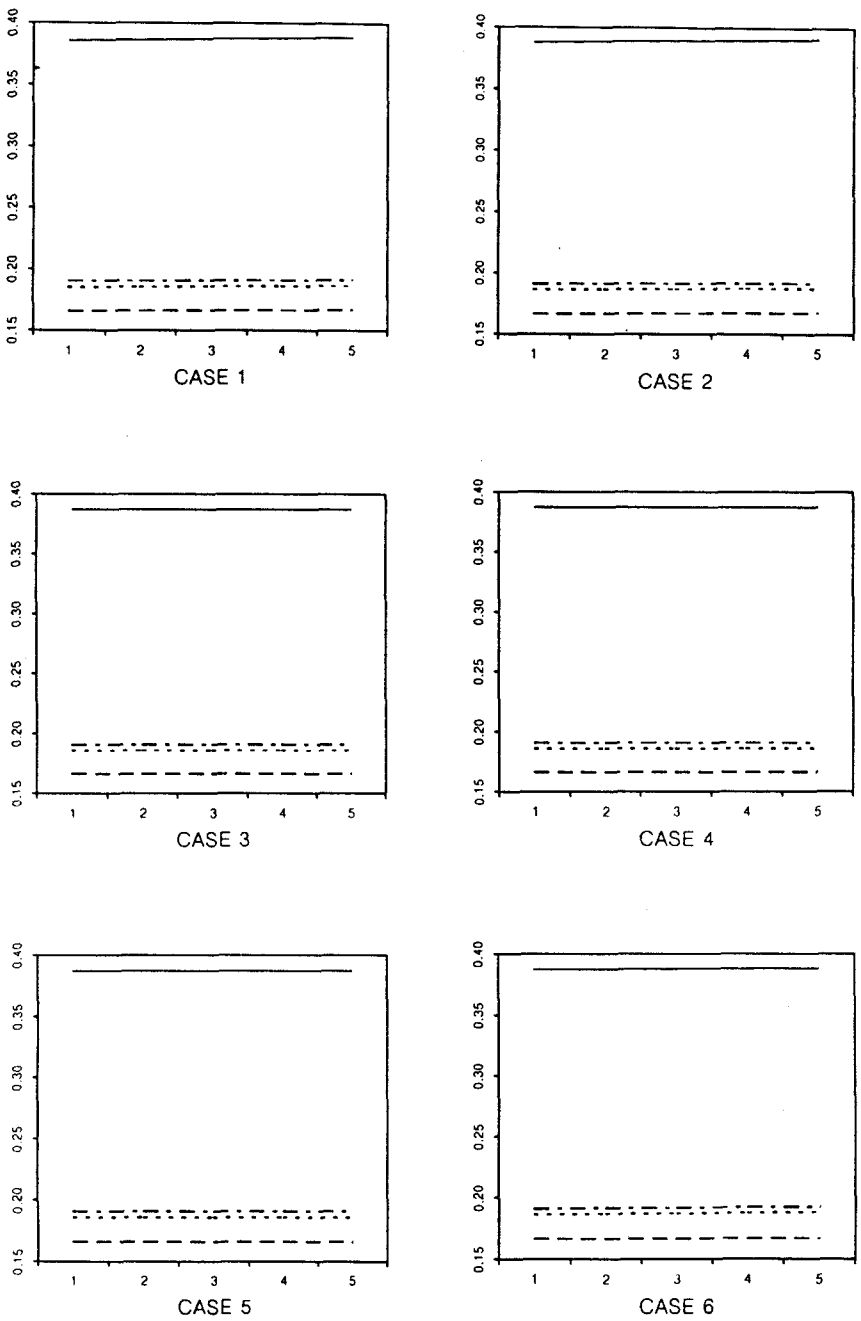


Figure 2: Absolute value; graphs of $\bar{\hat{\Lambda}}_{\cdot j}$

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In Figures 1 and 2, the averages of the absolute values in the column of $\hat{F}(\hat{F}_{.1}, .., \hat{F}_{.4})$ and $\hat{\Lambda}(\hat{\Lambda}_{.1}, .., \hat{\Lambda}_{.4})$ are plotted on the y-axis; the hyperparameter values we choose to change for the 6 distinct cases which are the small changes of the hyperparameters of the contaminated priors: $(\Lambda_1, B_1, H_1, \epsilon)$ are plotted on the x-axis. In these figures, a horizontal line graph(zero slope) reflects an insensitivity or robustness of that parameter to the variation, whereas to the variations, that is, a lack of robustness. We see that only cases 1 and 2 show a lack of robustness, and those cases involve a variation in Λ_0. The other cases involving the other hyperparameters show a strong robustness.

In Table 2, we determined that a case was sensitive if the distance measures were not near zero. Consequently, from the evidence, we conclude that Bayes estimates in Factor Analysis are robust respect to a prior ϵ-contamination class, except for the variation with respect tp Λ_0, the location parameter. Therefore, great care should be taken in assessing this hyperparameter.

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